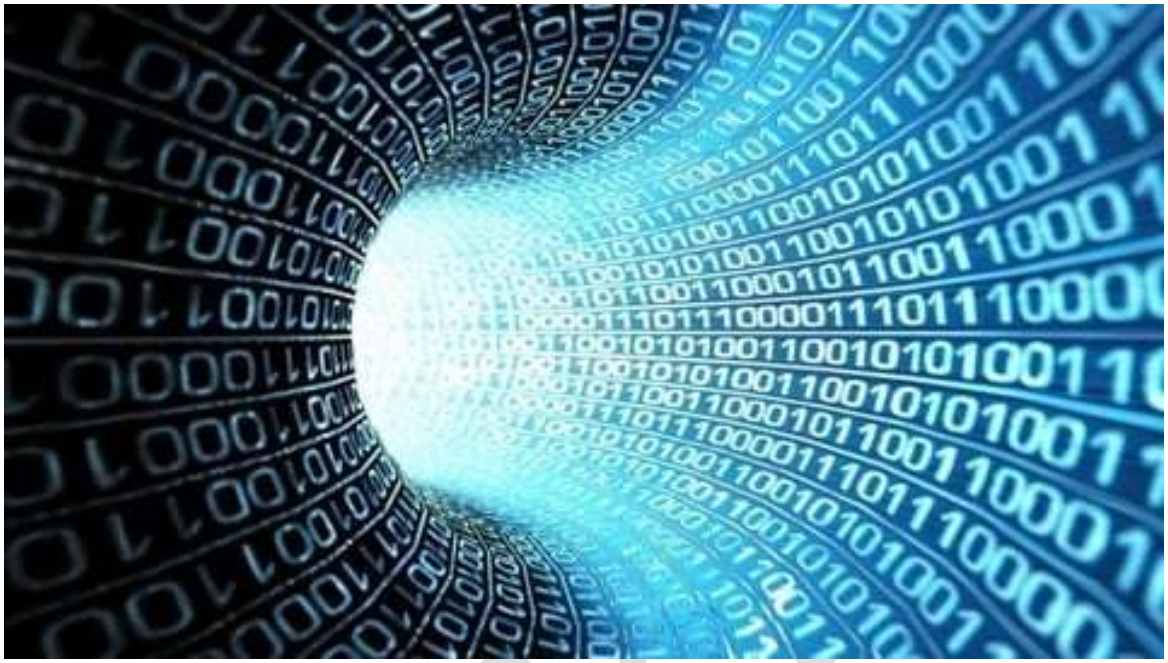
Networking

Basic Clearance of Networking Concepts



This module explains how data is created, structured, transmitted, routed, and secured across networks. It covers layered models (OSI/TCP-IP), encapsulation, addressing, transport behavior, and real-world protocols like DNS, HTTP, FTP, and email systems. It further demonstrates how insecure protocols are upgraded using TLS, SSH, and VPN. The focus is on **internal working + command-level interaction + packet-level understanding**.

This documentation and learning approach is strongly inspired by the structured, hands-on content provided by TryHackMe, which significantly contributed to practical understanding of Networking concepts.

DOCUMENTED BY

Kaustubh Ghosalkar aka "WEIRDO"

LINKEDIN GITHUB

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I. Networking Concepts

1. The OSI (Open Systems Interconnection) Model

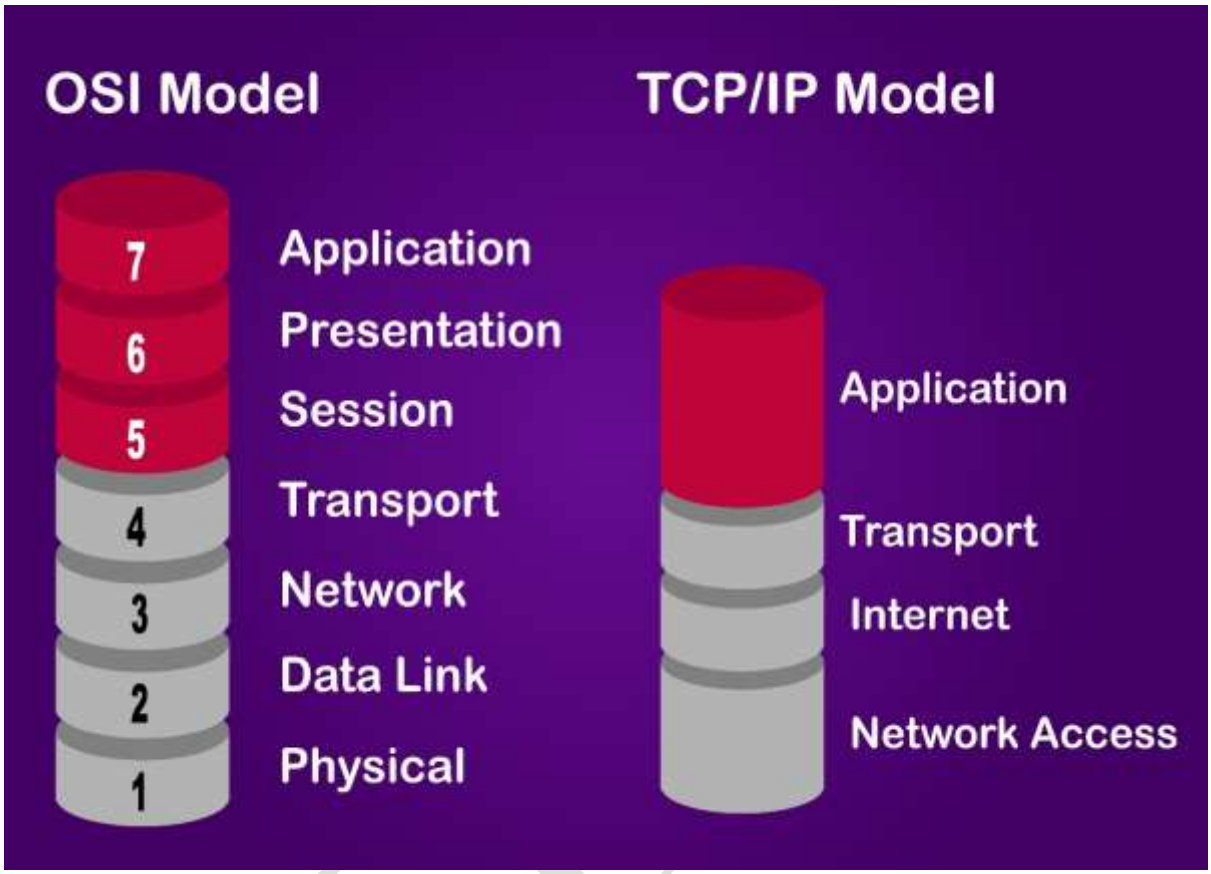
The **OSI Model** is a theoretical framework developed by the **ISO** to standardize how computer systems communicate over a network. It consists of **seven distinct layers**, numbered from the bottom (Layer 1) to the top (Layer 7).

Layer Number	Layer Name	Primary Function	Protocol Examples
Layer 7	**Application**	User-level services and network interfaces.	**HTTP, FTP, DNS, SMTP**
Layer 6	**Presentation**	Data encoding, encryption, and compression.	**SSL/TLS, MIME, JPEG**
Layer 5	**Session**	Session management and synchronization.	**NFS, RPC**
Layer 4	**Transport**	End-to-end communication and flow control.	**TCP, UDP**
Layer 3	**Network**	Logical addressing and path routing.	**IP, ICMP, IPSec**
Layer 2	**Data Link**	Communication between nodes on the same segment.	**Ethernet, WiFi (802.11)**
Layer 1	**Physical**	Physical transmission of bits via hardware.	**Fiber, Copper, Radio**

- **Mnemonic:** "**P**lease **D**o **N**ot **T**hrow **S**pinach **P**izza **A**way" (Physical to Application).
- **Layer 2 (Data Link):** Uses **MAC (Media Access Control)** addresses (6 bytes, hexadecimal). The first three bytes identify the vendor.
- **Layer 3 (Network):** Handles **Routing** across different network segments.

2. The TCP/IP Model

The **TCP/IP Model** (created by the DoD) is the practical implementation used in modern networking. It groups several OSI layers together into a **four-layer** (or sometimes five-layer) architecture.



- **Application Layer:** Combines OSI Layers 5, 6, and 7.
- **Transport Layer:** Maps directly to OSI Layer 4.
- **Internet Layer:** Maps to OSI Layer 3 (Network).
- **Link Layer:** Maps to OSI Layer 2 (Data Link). (Some models include **Layer 1** as the Physical layer).

3. IP Addressing and Subnetting

Every host requires a **unique identifier** to communicate. **IPv4** addresses are 32-bit addresses divided into **four octets** (0-255).

Key Components:

- **Network Address:** The first address in a subnet (e.g., **192.168.1.0**).
- **Broadcast Address:** Used to target all hosts on the subnet (e.g., **192.168.1.255**).
- **Subnet Mask:** Defines which part of the address is the network vs. the host (e.g., **/24** or **255.255.255.0**).

Private IP Address Ranges (RFC 1918):

These addresses are for internal use and are not routable on the public internet.

1. **10.0.0.0 – 10.255.255.255**
2. **172.16.0.0 – 172.31.255.255**
3. **192.168.0.0 – 192.168.255.255**

Command Section: Network Configuration

Command	System	Description
ipconfig	**Windows**	Displays IP address, subnet mask, and gateway.
ifconfig	**Linux/Unix**	Traditional command to view network interface details.
ip a s	**Linux**	Modern command (**ip address show**) for interface data.

4. Transport Protocols: TCP vs. UDP

These protocols operate at **Layer 4** to facilitate communication between specific processes using **Port Numbers** (1 - 65535).

UDP (User Datagram Protocol)

- **Connectionless:** Does not establish a formal connection before sending.
- **Unreliable:** No guarantee of delivery (Best Effort).
- **Speed:** Faster than TCP due to low overhead.
-

TCP (Transmission Control Protocol)

- **Connection-oriented:** Requires a formal setup.
- **Reliable:** Uses **Sequence Numbers** and **Acknowledgements (ACK)** to ensure data arrives correctly.
- **Three-Way Handshake:**
    1. **SYN:** Client sends a synchronization request.
    2. **SYN-ACK:** Server acknowledges and sends its own sync request.
    3. **ACK:** Client acknowledges the server's response.

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5. Encapsulation

Encapsulation is the process where each layer adds its own header to the data as it moves down the stack.

1. **Application Layer:** Generates raw **Data**.
2. **Transport Layer:** Adds TCP/UDP header => **Segment/Datagram**.
3. **Network Layer:** Adds IP header => **Packet**.
4. **Link Layer:** Adds MAC header and trailer => **Frame**.

6. Practical Communication with Telnet

Telnet is a legacy protocol used to communicate with remote systems over **TCP**. It is useful for testing connectivity to specific ports.

Command Examples:

- **Connect to Echo Server (Port 7):**

telnet [IP_ADDRESS] 7

- **Connect to Daytime Server (Port 13):**

telnet [IP_ADDRESS] 13

- **Connect to Web Server (Port 80):**

telnet [IP_ADDRESS] 80

Manual HTTP Request Example:

```
Bash
GET / HTTP/1.1
Host: telnet.thm
[Press Enter Twice]
```

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II. Networking Essentials

1. DHCP: Dynamic Host Configuration Protocol

DHCP is an **Application-layer protocol** used to automatically assign network settings to devices. It eliminates the need for manual configuration and prevents **IP address conflicts**.

- **Protocol Details:** Relies on **UDP**.
- **Port Numbers:** Server listens on **Port 67**; Client sends from **Port 68**.
- **Essential Configurations Provided:** IP Address, Subnet Mask, Gateway (Router), and DNS Server.

The DORA Process

DHCP operations follow four specific steps:

Step	Message	Description
1	**Discover**	Client broadcasts (0.0.0.0 to 255.255.255.255) to find a DHCP server.
2	**Offer**	Server responds with an available IP address and network settings.
3	**Request**	Client broadcasts a message to accept the specific offered IP.
4	**Acknowledge**	Server confirms the assignment and completes the configuration.

2. ARP: Address Resolution Protocol

ARP bridges **Layer 3 (Network)** and **Layer 2 (Data Link)** by finding the **MAC address** associated with a known **IP address**. Communication on a local segment (Ethernet/WiFi) requires the destination MAC address to build a frame.

ARP Mechanism

1. **ARP Request:** A broadcast frame sent to **ff:ff:ff:ff:ff:ff** asking "Who has [Target IP]?".
2. **ARP Reply:** The target device responds directly (unicast) with its MAC address.

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3. ICMP: Internet Control Message Protocol

ICMP is used for **network diagnostics** and error reporting. It does not carry user data but provides feedback on communication issues.

Key Diagnostic Commands

A. Ping

Tests connectivity to a target and measures **Round-Trip Time (RTT)**.

- **ICMP Type 8:** Echo Request (Sent by the source).
- **ICMP Type 0:** Echo Reply (Sent by the target).

B. Traceroute (tracert on Windows)

Discovers the path (routers) between the source and destination.

- **Logic:** Uses the **Time-to-Live (TTL)** field in the IP header.
- **Mechanism:** Every router decrements TTL by 1. When TTL reaches 0, the router drops the packet and sends an **ICMP Type 11 (Time Exceeded)** message back to the source.

Command Section: ICMP Tools

Command	System	Example	Description
ping	Linux/Windows	ping 8.8.8.8 -c 4	Sends 4 echo requests to test connectivity.
traceroute	Linux	traceroute google.com	Maps the hop-by-hop path to the host.
tracert	Windows	tracert google.com	Windows version of the traceroute tool.

4. Routing Protocols

Routing is the process of selecting paths across a network. Different protocols use various algorithms to determine the "best" path.

Protocol	Name	Type	Key Characteristic
OSPF	Open Shortest Path First	Link-State	Uses a map of the network to find the most efficient path.
EIGRP	Enhanced Interior Gateway Routing Protocol	Hybrid	**Cisco Proprietary**; considers bandwidth and delay.
BGP	Border Gateway Protocol	Path-Vector	The primary protocol of the **Internet** (connects ISPs).
RIP	Routing Information Protocol	Distance-Vector	Simple protocol; chooses paths based on the fewest **hops**.

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5. NAT: Network Address Translation

NAT allows a **private network** to use a single **public IP address** for Internet access. This was developed to solve **IPv4 address exhaustion**.

- **Mechanism:** The router maintains a **translation table** that maps internal (Private IP + Port) to external (Public IP + Port).
- **Benefit:** Enables thousands of simultaneous connections (up to approximately **65,000** per public IP) using unique port numbers.

➢ Conclusion and Cheatsheet

Conclusion

This module establishes the "glue" of networking. While the OSI model provides the theory, DHCP, ARP, and NAT provide the practical means for devices to join networks, locate peers, and share limited resources. ICMP serves as the primary toolset for verifying these connections and identifying where failures occur in the transmission path.

Fast Revision Cheatsheet

- **DHCP Ports:** 67 (Server), 68 (Client).
- **DORA:** Discover, Offer, Request, Acknowledge.
- **ARP:** Translates IP to MAC.
- **Broadcast MAC:** ff:ff:ff:ff:ff:ff.
- **ICMP Type 8/0:** Ping Request/Reply.
- **ICMP Type 11:** Time Exceeded (used by Traceroute).
- **TTL:** Decremented by routers; causes ICMP Type 11 when it hits 0.
- **Private IP Ranges:** 10.x.x.x, 172.16.x.x - 172.31.x.x, 192.168.x.x.
- **NAT:** Maps private IPs to one public IP to save address space.

III. Network Core Protocols

1. DNS: Domain Name System

DNS serves as the "phonebook" of the internet, mapping human-readable domain names to machine-readable **IP addresses**. It operates at **Layer 7** (Application Layer) and typically uses **UDP Port 53** (falling back to **TCP Port 53** if needed).

Core DNS Record Types

- **A Record:** Maps a hostname to an **IPv4 address**.
- **AAAA Record:** Maps a hostname to an **IPv6 address**.
- **CNAME Record:** Maps a domain name to another **Canonical Name** (alias).
- **MX Record:** Identifies the **Mail Exchange** server responsible for the domain.

Command Section: DNS Investigation

Command	Example	Description
nslookup	nslookup www.google.com	Queries the default DNS server for the IP address of a host.
dig	dig example.com MX	(Linux) Advanced tool to query specific record types like MX or AAAA.

2. WHOIS Records

WHOIS is a protocol used to query databases that store the registered users or assignees of an internet resource, such as a domain name. It provides technical and contact information regarding the domain's **Registrant**.

- **Registration:** Domain owners must provide accurate contact details unless using **Privacy Protection** services.
- **Data Points:** Includes creation date, expiration date, registrar name, and contact emails.
-

Command Section: Domain Ownership

Command	Example	Description
whois	whois tryhackme.com	Displays registration details and expiry dates for the domain.

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3. HTTP and HTTPS: Web Protocols

Hypertext Transfer Protocol (HTTP) is the foundation of data communication for the World Wide Web. **HTTPS** adds a layer of security via encryption.

Common HTTP Methods

- **GET:** Retrieves data (HTML, images) from a server.
- **POST:** Submits data to a server (forms, uploads).
- **PUT:** Updates or creates a resource on the server.
- **DELETE:** Removes a specified resource.

Port Standards

- **HTTP: TCP Port 80**.
- **HTTPS: TCP Port 443**.

4. FTP: File Transfer Protocol

FTP is designed specifically for efficient bulk file transfers. It uses **TCP Port 21** for control commands but establishes a separate connection for data transfer.

FTP Command Set

- **USER / PASS:** Authentication credentials.
- **LIST:** Displays files in the current directory.
- **RETR: Retrieve** (download) a file.
- **STOR: Store** (upload) a file.
- **Anonymous Login:** Many servers allow login using the username "anonymous" with no password.

5. Email Protocols: SMTP, POP3, and IMAP

Email relies on three distinct protocols to handle the flow of messages from the sender to the recipient's local device.

SMTP (Simple Mail Transfer Protocol)

- **Function:** Used for **sending** emails.
- **Port: TCP Port 25**.
- **Key Commands:** HELO, MAIL FROM, RCPT TO, DATA, and . (on a new line to end the message).

POP3 (Post Office Protocol v3)

- **Function:** Used for **retrieving** emails.
- **Port: TCP Port 110**.
- **Behavior:** Usually **downloads and deletes** the message from the server.
- **Key Commands:** USER, PASS, STAT (status), LIST, RETR.

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IMAP (Internet Message Access Protocol)

- **Function:** Used for **synchronizing** emails across multiple devices.
- **Port: TCP Port 143**.
- **Behavior:** Keeps messages on the server, allowing for folder syncing and "read" status updates across clients.
- **Key Commands:** LOGIN, SELECT, FETCH.

➢ Conclusion and Cheatsheet

Conclusion

This module illustrates that while modern internet usage is wrapped in graphical interfaces, the underlying infrastructure relies on simple, text-based command exchanges. By mastering protocols like **DNS** for resolution, **HTTP** for web access, and **SMTP/IMAP** for communication, network administrators and security analysts can effectively debug traffic and identify misconfigurations at the Application Layer.

Fast Revision Cheatsheet

Service	Protocol	Default Port	Key Purpose
Directory	**DNS**	**53 (UDP/TCP)**	Name to IP Resolution.
Web	**HTTP**	**80 (TCP)**	Non-encrypted web traffic.
Web	**HTTPS**	**443 (TCP)**	Encrypted web traffic.
Files	**FTP**	**21 (TCP)**	High-speed file transfer.
Mail Send	**SMTP**	**25 (TCP)**	Transferring mail between servers.
Mail Get	**POP3**	**110 (TCP)**	Download/Delete retrieval.
Mail Sync	**IMAP**	**143 (TCP)**	Multi-device synchronization.
Remote	**TELNET**	**23 (TCP)**	Remote terminal communication.

IV. Networking Secure Protocols

1. TLS: Transport Layer Security

TLS is a cryptographic protocol designed to provide secure communication over a network. It ensures that data cannot be read (**Confidentiality**) or altered (**Integrity**) by unauthorized parties.

- **Evolution:** TLS 1.0 (1999) was an upgrade to **SSL 3.0**. The current standard is **TLS 1.3** (2018).
- **Core Goals:**
  - **Confidentiality:** Prevents eavesdropping (packet sniffing).
  - **Integrity:** Prevents data modification during transit.
  - **Authenticity:** Ensures the client is talking to the legitimate server.

Digital Certificates and Trust

To identify itself, a server uses a **TLS Certificate**.

- **CSR (Certificate Signing Request):** An administrator sends this to a **CA (Certificate Authority)**.
- **CA (Certificate Authority):** A trusted third party that signs and issues the certificate.
- **Self-Signed Certificates:** Created by the user. These provide encryption but **cannot prove authenticity** and are often flagged as untrusted by browsers.

2. HTTPS: Hypertext Transfer Protocol Secure

HTTPS is simply **HTTP over TLS**. While standard HTTP sends data in cleartext, HTTPS encrypts the entire Application Layer exchange.

Connection Sequence

1. **TCP Three-Way Handshake:** Establishes the initial connection.
2. **TLS Negotiation:** Server sends its certificate; client and server agree on encryption keys.
3. **Encrypted Application Data:** Standard HTTP requests (GET, POST) are sent through the secure tunnel.

3. Secure Email Protocols

Standard email protocols (SMTP, POP3, IMAP) are inherently insecure. Adding **TLS** provides the necessary security for login credentials and private messages.

Insecure Protocol	Port	Secure Protocol (over TLS)	Secure Port
SMTP	25	**SMTPS**	465 or 587
POP3	110	**POP3S**	995
IMAP	143	**IMAPS**	993

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4. SSH: Secure Shell

SSH was created to replace **TELNET** (Port 23) for remote system administration. It provides a secure, encrypted channel over an insecure network.

- **Primary Port: TCP Port 22**.
- **Features:**
  - **Secure Authentication:** Supports passwords, **Public Keys**, and Two-Factor Authentication.
  - **Tunneling:** Can route other protocols through an encrypted SSH tunnel.
  - **X11 Forwarding:** Allows running graphical applications from a remote server.

Command Section: SSH Usage

Command	Example	Description
ssh	ssh user@10.10.10.10	Connects to a remote host as a specific user.
ssh -X	ssh -X user@host	Enables **X11 Forwarding** for graphical apps.

5. Secure File Transfer: SFTP vs. FTPS

While both offer security, they rely on different underlying mechanisms.

Feature	SFTP (SSH File Transfer Protocol)	FTPS (FTP over TLS)
Underlying Protocol	**SSH**	**TLS**
Port	**22**	**990** (usually)
Setup	Easy (part of OpenSSH)	Complex (requires CA certificates)
Firewall	Friendly (single port)	Difficult (uses multiple connections)

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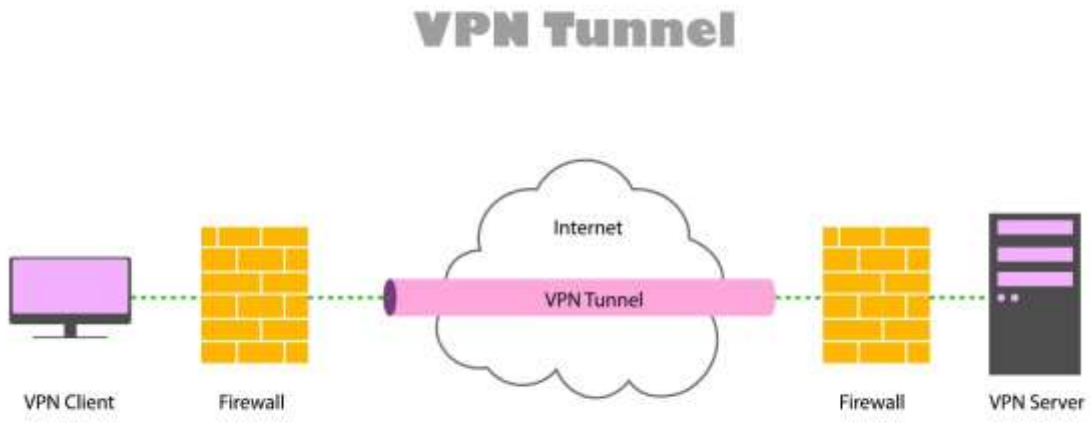
6. VPN: Virtual Private Network

A **VPN** creates a **Virtual** and **Private** connection over public infrastructure (the Internet). It uses a process called **Tunneling** to encapsulate and encrypt all traffic from a device or site.

- **Privacy:** Masks the user's real public IP address with the VPN server's IP.
- **Bypassing Restrictions:** Allows users to circumvent geographical censorship.
- **Corporate Use:** Connects remote branch offices to a main headquarters securely.



Conclusion and Cheatsheet

Conclusion

The transition from plaintext to secure protocols is essential for modern data protection. **TLS** provides the cryptographic foundation for web and email security, while **SSH** secures administrative access and file transfers. By utilizing these protocols and **VPN** technology, organizations can ensure that even when data travels over public, untrusted networks, its **confidentiality** and **integrity** remain uncompromised.

Fast Revision Cheatsheet

- **TLS 1.3:** The latest standard for secure transport.
- **HTTPS Port:** 443.
- **SSH Port:** 22 (Replaces Telnet Port 23).
- **SMTPS Ports:** 465 / 587.
- **IMAPS/POP3S Ports:** 993 / 995.
- **SFTP:** Uses SSH (Port 22); **FTPS** uses TLS (Port 990).
- **CA (Certificate Authority):** The "Trusted Third Party" that signs digital certificates.
- **Tunneling:** The core mechanism of VPNs to hide traffic.